

Probabilistic Robot Localization

Kalman Filter · Extended Kalman Filter · Particle Filter

TurtleBot3 Waffle — ROS 2 Jazzy — Gazebo

SPECIAL TASK 2510331009 · Particle Degeneration Analysis

AGENDA

- 01 System & Filter Overview
- 02 Experimental Setup & Landmark Correction
- 03 Q / R Sensitivity Study
- 04 **Special Task 2510331009 — Particle Degeneration**
- 05 Discussion, Status & Takeaways

Problem Statement

- Mobile robots must estimate their pose under **uncertain motion** and **noisy sensing**
- Three classical probabilistic state estimators: **KF**, **EKF**, **Particle Filter** — all implemented on one platform with identical inputs
- **Special Task 2510331009**: disable PF resampling and analyze particle degeneration

Platform	Framework	Language	State vector
TurtleBot3 Waffle	ROS 2 Jazzy · Nav2 · Gazebo	C++ (rclcpp, Eigen3) + Python 3	$\mu = [x, y, \theta]^T$ (odom frame)

System Architecture — ROS 2 Nodes

Single `ros2 launch` starts the full pipeline: Gazebo + Nav2 (0 s) → Filters + RViz + Logger (5 s) → AutoNav (10 s)

Node	Executable	Subscribes	Publishes
Kalman Filter	kf_node	/cmd_vel, /odom, /scan	/pose_kf
Extended KF	ekf_node	/cmd_vel, /odom, /scan	/pose_ekf
Particle Filter	pf_node	/cmd_vel, /odom, /scan	/pose_pf, /pf_particles, /n_eff
Data Logger	data_logger	/odom, /pose_kf, /pose_ekf, /pose_pf, /n_eff	CSV file
AutoNav	auto_nav	—	/initialpose, Nav2 goals

State vector (all filters): $\mu = [x, y, \theta]^T$ — reference frame: odom

Launch args: pf:=true resampling:=false sigma_process:=.. sigma_meas:=..

Filters at a Glance

	Kalman Filter	Extended KF	Particle Filter
Belief	Gaussian (μ, Σ)	Gaussian (μ, Σ)	N = 500 weighted particles
Prediction	Linearized B matrix cos/sin of θ — accepted approx.	Exact nonlinear $g(u, \mu)$ + Jacobian G — no linearization error	Sample $x_t^m \sim p(x_t u_t, x_{t-1}^m)$ Gaussian noise on v, ω
Correction	$C = I_3$ vs /odom + linearized landmark Jacobian	$H = I_3$ vs /odom + landmark Jacobian H_j	Importance weights $w p(z x^m)$; odom + landmark
Reference	Thrun Table 3.1	Thrun Table 3.3 / 7.2	Thrun Table 4.3 / 4.4

All three filters share the same landmark-based correction → differences come purely from prediction & belief representation

Shared Landmark-Based Correction

- One **shared landmark map** (landmark_map.hpp) for all three filters — identical correction input, fair comparison
- **Range-only** measurement model — expected bearing from pose estimate indexes into /scan for measured range
- **Gating:** $|r_meas - r_exp| \leq 0.30 \text{ m}$ accepted as valid; $\sigma_r = 0.12 \text{ m}$
- **Two-stage correction:** (1) full-state update vs /odom, then (2) sequential 1D range update per visible landmark

Measurement model

$$h(\mu, m_j) = \|m_j - [x, y]\|$$

Gate & noise

$$LM_GATE_M = 0.30 \text{ m}$$

$$LM_SIGMA_R = 0.12 \text{ m}$$

Experimental Setup

- Spawn pose: $(-1.96, -0.5, 0^\circ)$ in map frame = odom origin
- AutoNav drives a fixed **8-waypoint loop** across 4 unique positions — ~3.6 m diameter
- Ground truth proxy: `/odom` (Gazebo) — same caveat applies equally to all filters
- Identical route, timing, and sensor stream fed to KF, EKF, and PF on every run

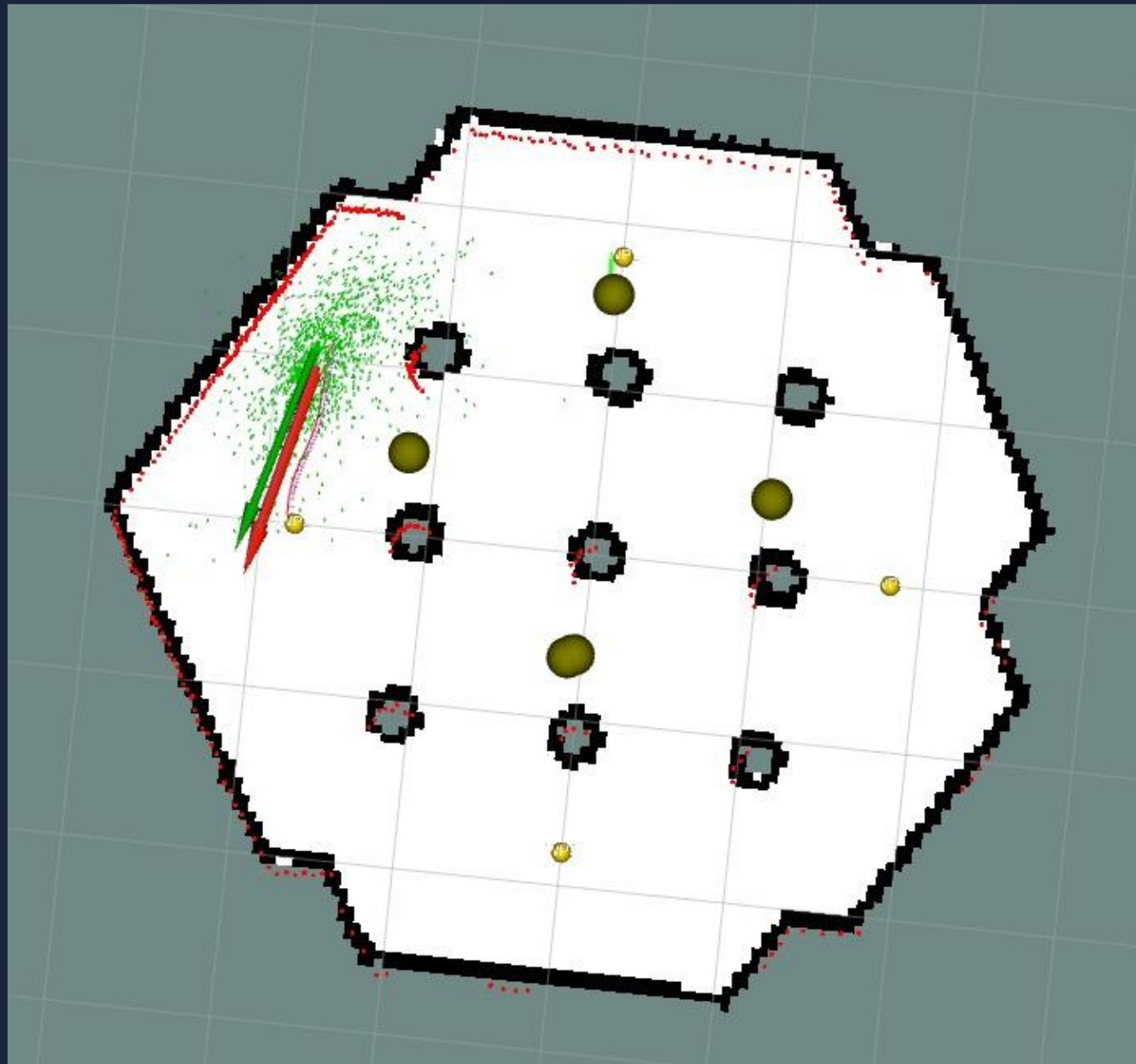
WAYPOINTS

WP1/6 N (0.0, 1.8)	WP2/5/8 W (-1.8, 0.0)
WP3/7 S (0.0, -1.8)	WP4 E (1.8, 0.0)

DEFAULT PARAMETERS

σ_{process}	0.01
σ_{meas}	0.10
num_particles	500

RViz2 Visualization

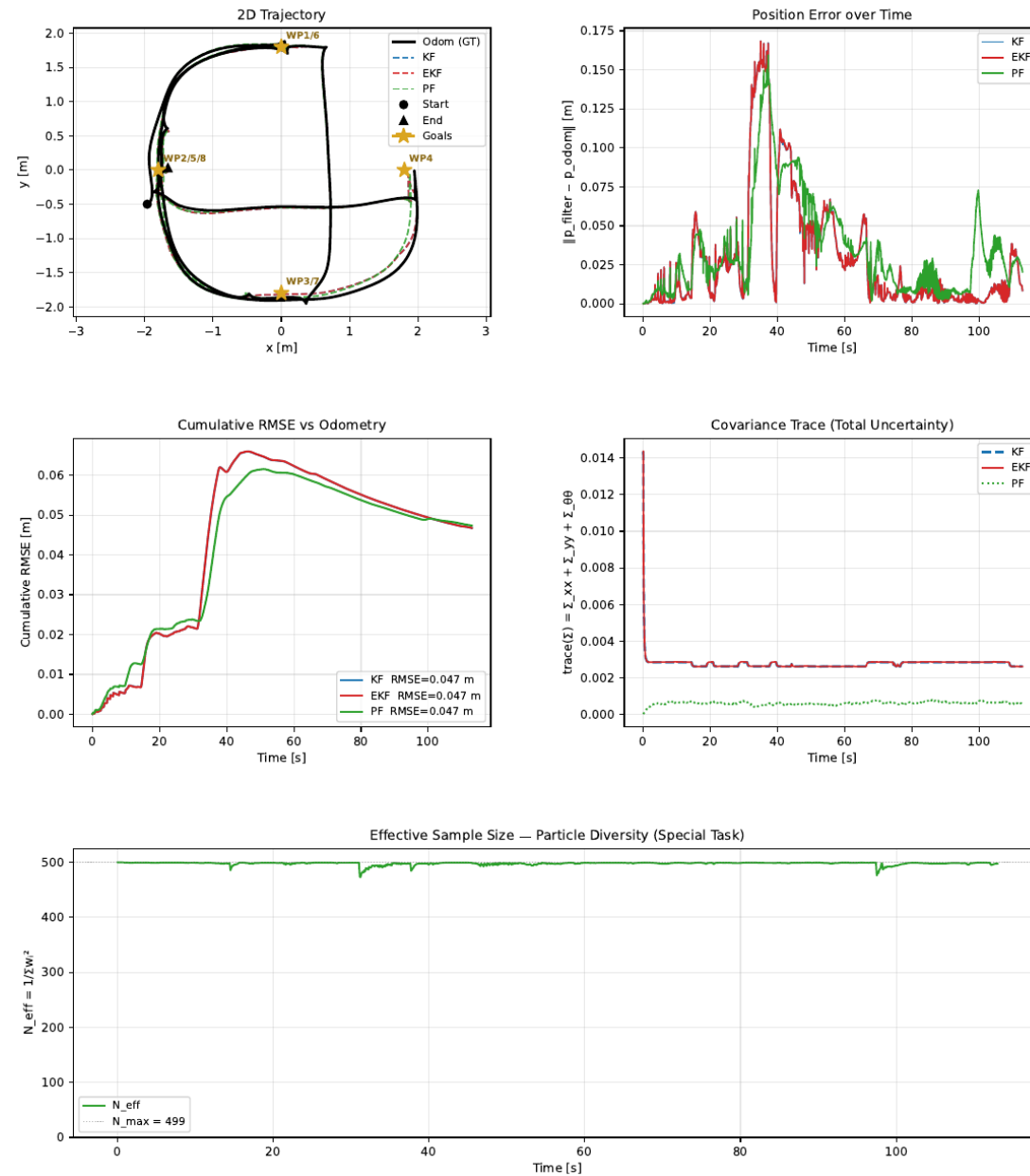


- Particle cloud — PF belief distribution
- Laser scan returns
- Landmark markers (yellow spheres)
- Pose estimate arrow + costmap boundary

Captured during exploratory run — included to satisfy the RViz2 visualization requirement

Baseline Filter Comparison (Resampling ON)

Probabilistic Robot Lab — KF / EKF / PF Filter Comparison
Ground truth: /odom (Gazebo)



RMSE vs /odom

KF = **0.047 m** EKF = **0.047 m** PF = **0.047 m**

- All three trajectories closely overlap with /odom ground truth across the full 8-waypoint loop
- PF covariance trace visibly lower — particle-based estimate tends to underestimate once particles cluster
- $N_{\text{eff}}/N \approx 1.0$ throughout — healthy particle diversity maintained

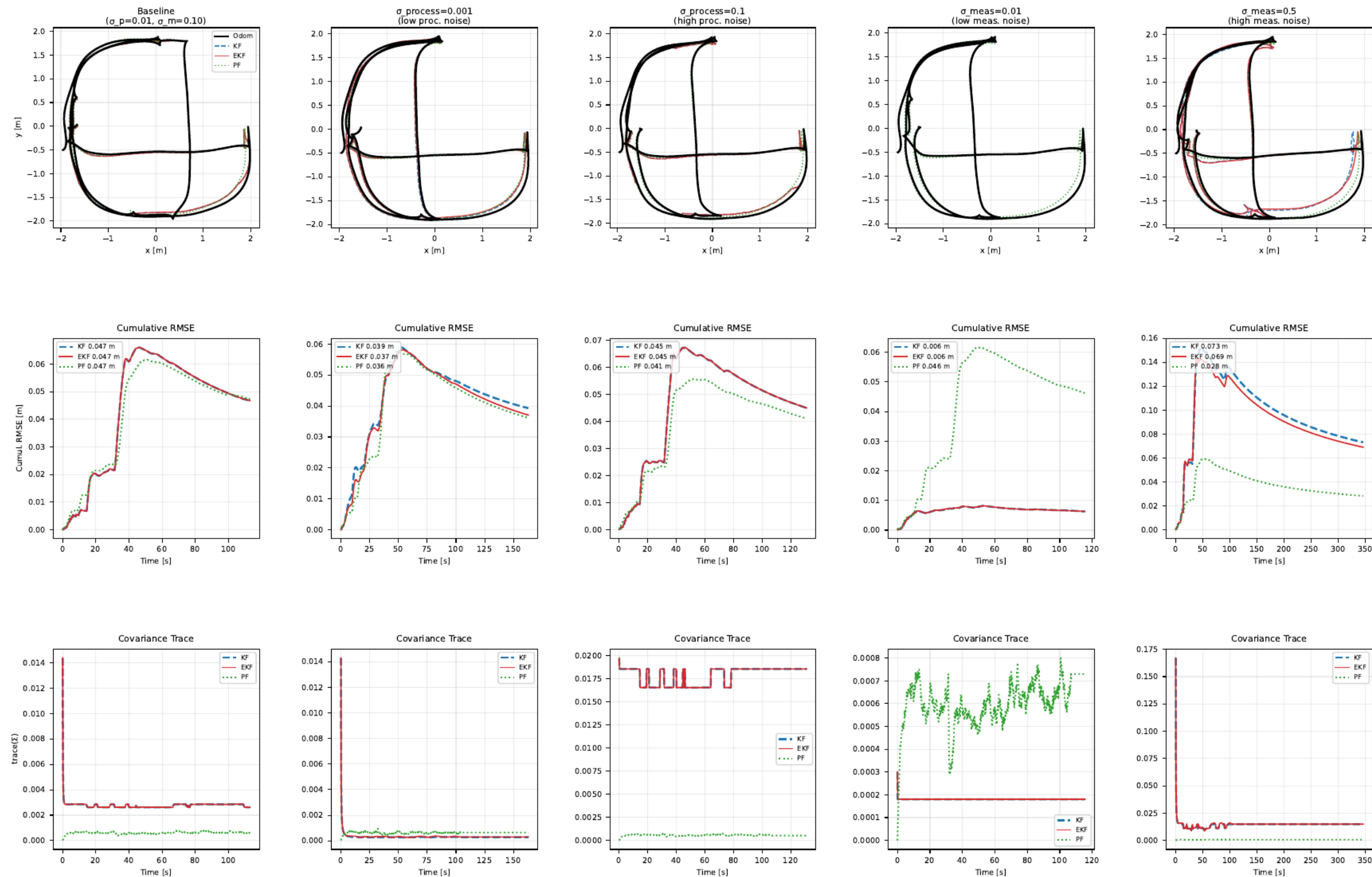
Q / R Sensitivity Study

- Vary **process noise** (Q, model confidence) and **measurement noise** (R, sensor trust) independently
- **5 runs total:** Baseline + low/high σ_{process} + low/high σ_{meas} — same route and landmark map throughout

Run	σ_{process}	σ_{meas}
Baseline	0.01	0.10
Low proc. noise	0.001	0.10
High proc. noise	0.10	0.10
Low meas. noise	0.01	0.01
High meas. noise	0.01	0.50

Q / R Results — Trajectories · RMSE · Covariance

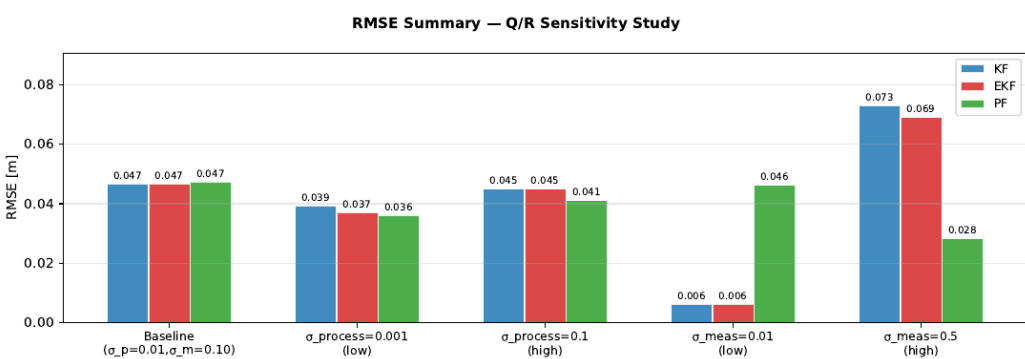
Q / R Sensitivity Study — KF / EKF / PF
Effect of process noise (σ_{process}) and measurement noise (σ_{meas})



Q / R Sensitivity — Summary

Run	σ_{proc}	σ_{meas}	KF	EKF	PF
Baseline	0.01	0.10	0.047	0.047	0.047
Low σ_{process}	0.001	0.10	0.039	0.037	0.036
High σ_{process}	0.10	0.10	0.045	0.045	0.041
Low σ_{meas}	0.01	0.01	0.006	0.006	0.046
High σ_{meas}	0.01	0.50	0.073	0.069	0.028

- **$\sigma_{\text{meas}}=0.01$:** KF/EKF RMSE collapses to 0.006 m — circularity artifact (filter trusts /odom almost unconditionally)
- **$\sigma_{\text{meas}}=0.5$:** KF/EKF rise to ~0.07 m; PF stays at 0.028 m — non-parametric PF is more robust to high measurement uncertainty



SPECIAL TASK 2510331009

Disabling PF Resampling & Analyzing Particle Degeneration

What happens to particle diversity and filter robustness when low-weight particles are never replaced?

Special Task: Definition

- **Task 2510331009:** "Run the PF with reduced or no resampling" — implemented as a full on/off toggle via ROS 2 parameter (`resampling:=false`)
- Resampling is the textbook mechanism that prevents the **degeneracy problem** in Sequential Importance Sampling (SIR) filters
- Without it, all probability mass is expected to **concentrate on a single particle** — weight collapse

Research question

"What happens to particle diversity and filter robustness when low-weight particles are never replaced?"

Systematic Resampling (Thrun Table 4.4)

- 1 Build the cumulative weight sum over all N particles
- 2 Draw a single random offset $r \sim U[0, 1/N)$
- 3 For $i = 0..N-1$: pick the particle at cumulative position $r + i/N$

Result: high-weight particles duplicated; low-weight dropped; all weights reset to $1/N$ — $O(N)$, low-variance

- Triggered every correction step when `resampling = true`

```
# Thrun (2006) Table 4.4
```

```
c[1] = w[1]
for i = 2..N:
    c[i] = c[i-1] + w[i]
```

```
r ~ U[0, 1/N)
```

```
for i = 1..N:
    u = r + (i-1)/N
    pick j: c[j] ≥ u
```

Particle Degeneration & N_eff

- Without resampling, particle weights are **only multiplied** — **never reset** → diverge monotonically over time
- Particles with even slightly lower likelihood are **exponentially suppressed** and never replaced
- N_eff computed **before** the resampling decision → reflects true particle health regardless of whether resampling runs

$$N_{\text{eff}} = 1 / \sum$$

$$\frac{w_i^2}{\sum w_i^2} \in [1, N]$$

$N_{\text{eff}} \rightarrow N$ **Uniform weights**
 (ideal)
 $N_{\text{eff}} \rightarrow 1$ **Single particle dominates**
 (full degeneration)

Special Task — Experiment Design

- **Two full runs** on the identical 8-waypoint route: Resampling ON (baseline) vs. OFF (Special Task)
- Identical Q/R defaults, landmark map, and $N = 500$ particles for both runs
- `/n_eff` and `/pose_pf` logged throughout by `data_logger` — timestamped CSV per run
- Compared via `plot_n_eff.py` : N_{eff}/N over time + cumulative RMSE, both runs overlaid on the same axes

Resampling ON (baseline)

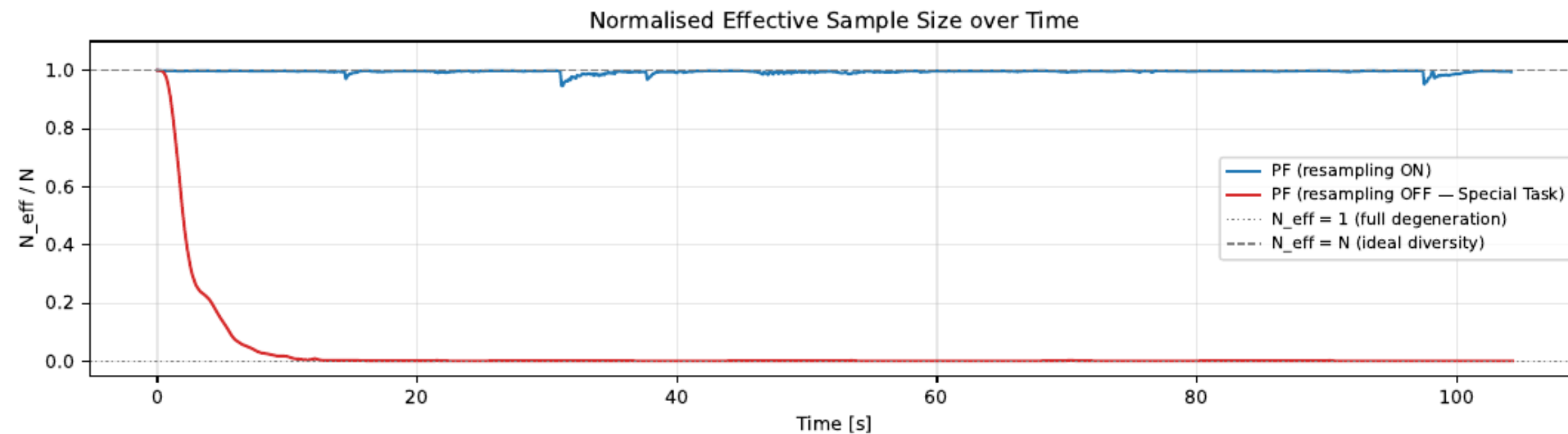
```
./run_experiments.sh --with-resampling
```

Resampling OFF (Special Task)

```
./run_experiments.sh --without-resampling
```

Results — N_{eff} Collapse

Special Task 2510331009 — Particle Degeneration Analysis
 $N_{\text{eff}} = 1/\sum w_i^2 \in [1/N, 1]$ (normalised to $[0, 1]$)

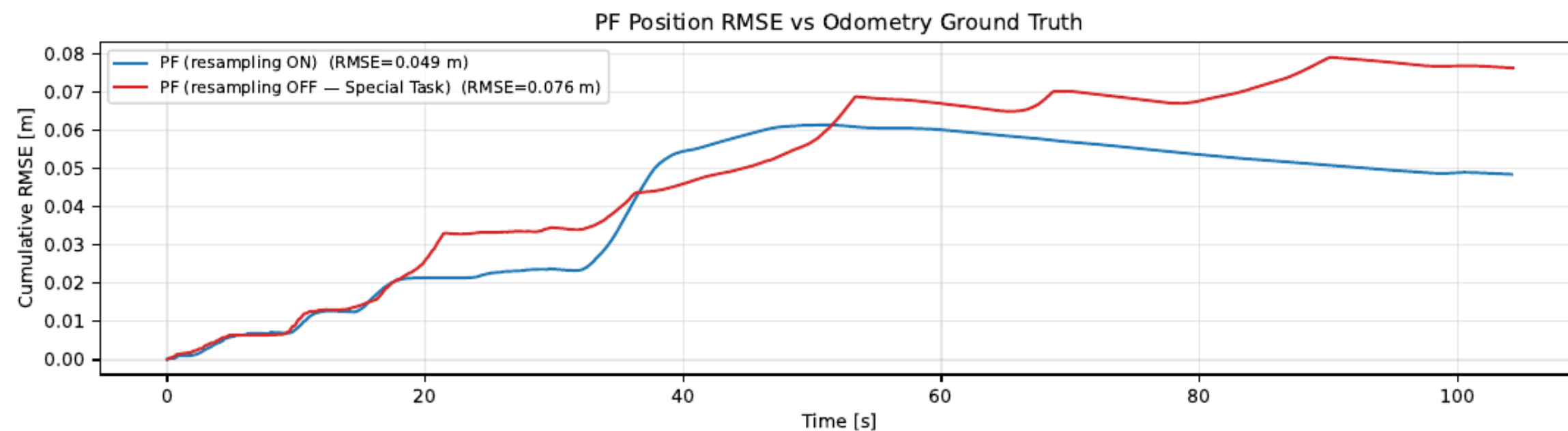


Resampling ON

$N_{\text{eff}}/N \approx 1.0$ throughout the entire ~110 s run, with brief dips at landmark corrections

Resampling OFF

N_{eff}/N collapses from ≈ 1.0 to ≈ 0.0 within **~20 seconds** — numerical zero for the rest of the run



Matches the theoretical prediction: without resampling, degeneration is **fast and irreversible**

Results — RMSE & Key Insight

Resampling ON

0.049 m

cumulative RMSE

Resampling OFF

0.076 m

cumulative RMSE (+55%)

- The single surviving particle keeps integrating odometry — RMSE alone still looks **"reasonable"** after full degeneration
- The filter has **lost its uncertainty representation** — cannot express ambiguity or recover from drift

KEY INSIGHT

RMSE is an **insufficient diagnostic** on its own.

N_{eff} is required to detect degeneration — a collapsed PF still reports a confident-looking pose and covariance.

Discussion & Interpretation

- Resampling trades short-term particle diversity for **long-term robustness** — classic bias/variance tradeoff in SIR filters
- Without resampling, the PF degrades into a **de-facto single-hypothesis filter** , losing multi-modality and recovery from ambiguous data association
- **Practical risk:** a degenerated filter still reports a confident-looking pose and covariance — failure goes undetected without explicitly monitoring N_{eff}
- Connects to Thrun (2006), Ch. 4.3: the *degeneracy problem* is the textbook motivation for resampling in SIR filters

Limitations

- **Single run per condition** — exact degeneration timing depends on the specific random noise realization
- **Binary on/off only** — no intermediate strategy tested (e.g. threshold-based resampling at $N_{\text{eff}}/N < 0.5$)
- **Ground truth is /odom** — the same circularity caveat applies as in the Q/R study
- **Short route (~110 s)** — degeneration effects over much longer missions remain untested

Possible extension: threshold-based resampling (resample only when $N_{\text{eff}}/N < 0.5$) to study an intermediate strategy

Implementation & Experiment Status

	KF	EKF	PF
Prediction step	✓	✓	✓
Correction (odom)	✓	✓	✓
Correction (landmarks)	✓	✓	✓
Full 3×3 covariance	✓	✓	✓
RViz visualization	✓	✓	✓
Timestamp check (stale odom)	✓	✓	—
Tunable Q/R	✓	✓	✓

EXPERIMENTS COMPLETED

- ✓ Ground truth RMSE vs /odom
- ✓ Covariance trace visualization
- ✓ Landmark detection (shared map)
- ✓ Special Task: N_eff with vs. without resampling
- ✓ σ_{process} variation (4 runs)
- ✓ σ_{meas} variation (4 runs)

KEY TAKEAWAYS

KF, EKF, and PF perform near-identically under benign noise on this shared, landmark-augmented setup — differences emerge mainly under high measurement uncertainty, where PF is most robust

Resampling prevents particle degeneration — disabling it causes near-total weight collapse (N_{eff}/N : 1.0 $\rightarrow$ ~ 0.0) within ~ 20 seconds

RMSE alone does not reveal degeneration — the surviving particle can still track odometry well short-term; N_{eff} is an essential complementary diagnostic

References

- **Thrun, S., Burgard, W., Fox, D. (2006). *Probabilistic Robotics*. MIT Press.**

Table 3.1 – Kalman Filter Table 3.3 / 7.2 – EKF / EKF Localization Table 4.3 – Particle Filter (MCL)

Table 4.4 – Systematic Resampling

- **Nav2 Documentation**

`docs.nav2.org`

- **ROS 2 Jazzy Documentation**

`docs.ros.org/en/jazzy`

- **Source code**

`github.com/Raini20/PROLB_R`

Thank You

Questions?

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`github.com/Raini20/PROLB_R`